

II-KL – Filtered Cube27 Rees/Ext localization, seventh Gauss Fredholm, codimension-three germinal atlas, second native Golden H^5 closure, full order-89 repair and recurrence orders 96–105

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Abstract

II-KL freezes II-KK and preserves the append-only HT+NT discipline. The Cube27 augmentation filtration is now serialized by degree-one multiplication data in every associated-graded degree and the nonzero conductor extension class is audited under 2-localization and Yoneda pushout. A seventh Gauss trace/Fredholm coefficient is enclosed with Arb and the simple leading Riesz point yields a genuine reduced-resolvent existence certificate, without inventing a numerical third pressure response. Codimension three gains a seven-level regular germinal atlas but no cross-singular quotient identification. Golden closes a second native H^5/B^5 carrier, type 4 of order 32, through six elementary-abelian E_8 restrictions. The recurrence engine is upgraded to FLINT: the previously resource-gated Fibonacci order 89 is completed in full and orders 96–105 are scanned exhaustively. No Stone/Fibonacci project value is fabricated.

1 Cube27: filtered Rees data and the 2-primary Ext class

Let I be the dimension-augmentation ideal. The low pieces remain

$$\begin{aligned} \mathrm{gr}_I^0 R &\simeq \mathbb{Z}, & \mathrm{SNF}(I/I^2) &= (2, 2, 12), \\ \mathrm{SNF}(I^2/I^3) &= (2, 2, 2, 2, 2, 12), & \mathrm{SNF}(I^n/I^{n+1}) &= (2, 2, 2, 2, 2, 2, 12) \quad (n \geq 3). \end{aligned}$$

II-KL now places in one authority object the 15 multiplication maps $\mathrm{gr}^1 \rightarrow \mathrm{gr}^2$, the 15 maps $\mathrm{gr}^2 \rightarrow \mathrm{gr}^3$, and the two 15-operator families of the proven odd/even period-two tail. Thirty descent checks and 225 low-degree commutativity checks pass. Since the declared augmentation ideal is generated by these fifteen classes, this reconstructs multiplication by degree-one generators on every gr_I^n . No new polynomial-quotient presentation of the completed Rees algebra is asserted.

For the conductor pullback class $e = [2x] \in \mathrm{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/2, A_2)$, II-KL computes

$$\boxed{\dim_{\mathbb{F}_2} A_2/2A_2 = 13}$$

and an explicit detector λ with $\lambda(e) = 1$. The class survives at the 2-localization and vanishes after inverting 2 (hence at every odd localization). Pushing out by λ gives a nonzero

$$\epsilon \in \mathrm{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/2, \mathbb{Z}/2),$$

while

$$\boxed{\epsilon^2 = 0 \in \mathrm{Ext}_{\mathbb{Z}}^2(\mathbb{Z}/2, \mathbb{Z}/2)}$$

because $\mathbb{Z}/2$ has projective dimension one over $\mathbb{Z}$.

2 Gauss and codimension-three holonomy

Seven inverse branches have determinant -1 . Summing the $7^7 = 823543$ exact branch contributions and enclosing the positive complement yields

$$\boxed{0.259748216751965 < \mathrm{tr}(L^7) < 14.797324802762404}.$$

Newton’s identity then gives

$$-2.116968096441766 < d_7 < 1.1953466299708688.$$

The interval crosses zero, so no sign is claimed.

The leading eigenvalue at $t = 0$ is simple and isolated with Riesz projector P inherited on the declared Gauss carrier. For $Q = I - P$ the reduced resolvent

$$S = Q(I - L_0|_{QX})^{-1}Q$$

exists as a bounded operator. Hence the third perturbative identity

$$\lambda_3 = \ell(L_3h) + 3\ell(L_2Su_1) + 3\ell(L_1Su_2), \quad P'''(0) = \lambda_3 - 3\lambda_2\lambda_1 + 2\lambda_1^3$$

is now well-typed and exact. A numerical norm for S , a numerical $P'''(0)$ and absolute time-domain three-point cumulant summability remain **SEPARATED**.

At 192-bit Arb precision the two unique regular crossings are further enclosed by

$$c_{\sqrt{5}/4} \in [0.8723484253712013, 0.8735363026114282], \quad c_{\pi/6} \in [0.9223291900725781, 0.9235602912103968].$$

Seven regular levels $c = 0.84, 0.86, \dots, 0.96$ carry interval-certified germs

$$H_c = R_{u(c)}S_0T, \quad H_c^2 = 1.$$

The source-bound bad-stratum atlas remains the alternating C_8 incidence of four S^3 and four T^2 pieces. No source-bound cross-singular holonomy arrows are recovered; the actual compactified leaf groupoid remains **SEPARATED**.

3 Golden-Hodge: second native H^5/B^5 closure

Type 2 remains closed at $\dim H^5 = 14$. II-KL treats type 4, whose stabilizer has order 32. Sixteen elementary-abelian E_8 subgroups are found on the actual group. The candidate family consists only of restrictions of actual native products built from inherited native H^1, H^2, H^3 representatives. For each selected E_8 , the actual inhomogeneous coboundary block

$$B^5(E) = \text{im}(\delta : C^4(E; \mathbb{F}_2) \rightarrow C^5(E; \mathbb{F}_2))$$

is inserted as an independent block in one joint augmented rank calculation. After six subgroups and

$$196608 \text{ streamed } C^5(E_8) \text{ rows}$$

the direct-sum restriction image reaches exactly 34 independent candidate pivots. The independent abstract upper bound is also 34, therefore

$$\dim H_{\text{native}}^5(\text{type 4}) = 34, \quad \text{native displayed products span it : PASS.}$$

This is the second native degree-five closure. It is not extrapolated to types 0, 1, 3, 5, 6. The genuine free $\mathbb{F}_2[H]$ Wall degree-four producer still lacks a free basis, equivariant d_4 , twisting data and Φ_4 bytes and remains **SEPARATED**.

4 Recurrence: full Fibonacci order 89 and orders 96–105

The previous order-89 resource overrun belonged to the SymPy implementation path. II-KL switches the factor ABI to `python-flint nmod_poly.factor` plus `fmpz.factor`, with cached factorizations of $3^d - 1$. The nine old II-KK order-89 probes agree 9/9 with the new engine. The complete sweep now contains

$$88/88 \text{ tap values at } k = 89$$

with 84 distinct matrix orders; no full global antiperiod occurs, in accordance with the odd-order determinant theorem. The old timeout is therefore *not* an engine-independent invariant. Since 89 is nevertheless a Fibonacci number, it remains an explicitly typed structural coordinate for later order/value comparisons.

All tap values are also factorized for every $k = 96, \dots, 105$:

$$995 \text{ new exact factor records.}$$

On $k = 96, 98, 100, 102, 104$, the reflection theorem passes

$$124/124$$

applicable odd- v pairs. The fixed centers are

k	$v = k/2$	$\text{ord } C$	h_{anti}
96	48	384	192
98	49	392	196
100	50	400	200
102	51	408	204
104	52	416	208

so again $\text{ord } C = 4k$ and $h_{\text{anti}} = 2k$. The repaired/new collision quotient contains 237 classes, 68 cross-order, including four classes involving the now-complete order 89. No new record satisfies $\text{ord } C = k$ or $k + v$. The divisibility coordinates remain richer than any single linear order/value law and are kept internal to the recurrence carrier.

5 D_5 , Pisano, project gates and checkpoint

Fresh post-A1BI retrieval yields no continuous rank-three D_5 transition packet in the declared Library/GitHub scope. This is scoped retrieval evidence only; the continuous packet remains NT. The historical Pisano $p = 5$ lane still lacks a complete additive-character/self-dual-Haar/conductor/uniformizer/Tate byte packet, so its scoped obstruction is preserved. Stone still lacks project d, θ, K_Z, C and the framing anchor; Fibonacci still lacks an instantiated carrier/localization/cofinal-basis/trace-table packet. Hence

ABI-complete packets = 0, promoted values = 0, 37 project gates active.

The append-only checkpoint is

1432 PASS, 97 ZERO, 450 SEPARATED, 294 REFUTED, 87 NT,

675 PROVEN-MERGEABLE, 456 PROVEN-NOT-MERGEABLE.

The deep verifier closes 10 families with 1727 explicit assertions, 1376 witness records and 1137096 coverage units. Observational continuity advances to PASS_297; exhaustive historical replay remains NT_NOT_EXHAUSTIVE.

Safe successor.

II-KM - Cube27 2-local conductor-Ext module filtration and completed Rees exactness/periodic-module audit; Gauss eighth Fredholm trace/coefficient and certified reduced-resolvent norm or proof-grade numerical third-pressure enclosure; codimension-three cross-singular quotient/holonomy source ingestion with sharper Arb crossing localization; Golden third native H5/B5 closure or a trusted free F2[H]-Wall degree-four producer; post-A1BI/A1BJ D_5 transition-source ingestion; Pisano historical p5 additive-character/Haar/Tate packet acquisition; recurrence FLINT repair of full orders 91/93/95 plus full orders 106-115 and divisibility/collision theorem refinement; plus external Stone/Fibonacci ABI-complete packet execution.